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Trade and ethnolinguistic differences: A replication and extension

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Abstract

We revisit the role of trade on long-run inter-ethnic linguistic differences. Dickens (2022) hypothesized that neighboring languages are more similar when agriculture provides potential gains from *inter-ethnic* trade. Since his empirical approach confounds inter- and intra-ethnic trade, we replicate his main analysis using improved measures of potential inter-ethnic gains from trade. Our results confirm the role of trade in inter-ethnic linguistic differences while showing the difficulty of disentangling the role of inter- versus intra-ethnic trade. We also provide an open-source computational framework to replicate both sets of results, which others can use to produce original replicable economic research.

KEYWORDS
geographic information systems, inter-ethnic trade, intra-ethnic trade, linguistic distances, mapping, replication, trade

JEL CLASSIFICATION
F10, O10, Z10

1 | INTRODUCTION

A growing literature has identified ethnolinguistic groups and their characteristics as central to our understanding of contemporary economic development, conflict, and other social, political, and economic phenomena (Acemoglu et al., 2020; Alesina et al., 2003; Alesina & La Ferrara, 2005; Depetris-Chauvin & Özak, 2020a; Easterly & Levine, 1997; Michalopoulos & Papaioannou, 2013). In turn, this has prompted the search of deep historical roots that explain the differences between these groups and their languages (Depetris-Chauvin & Özak, 2020b; Galor et al., 2016, 2018; Michalopoulos, 2012).

In an important recent contribution to this literature, Dickens (2022) studies the effect of potential gains from *inter-ethnic* trade on linguistic distances. He hypothesizes that neighboring ethnolinguistic groups, which interacted more often due to historical *inter-ethnic* trade opportunities, became more culturally similar. To test this hypothesis, he

Abbreviations: CSI, Caloric Suitability Index; FAO, Food and Agriculture Organization of the United Nations; GAEZ, Global Agro-Ecological Zones.

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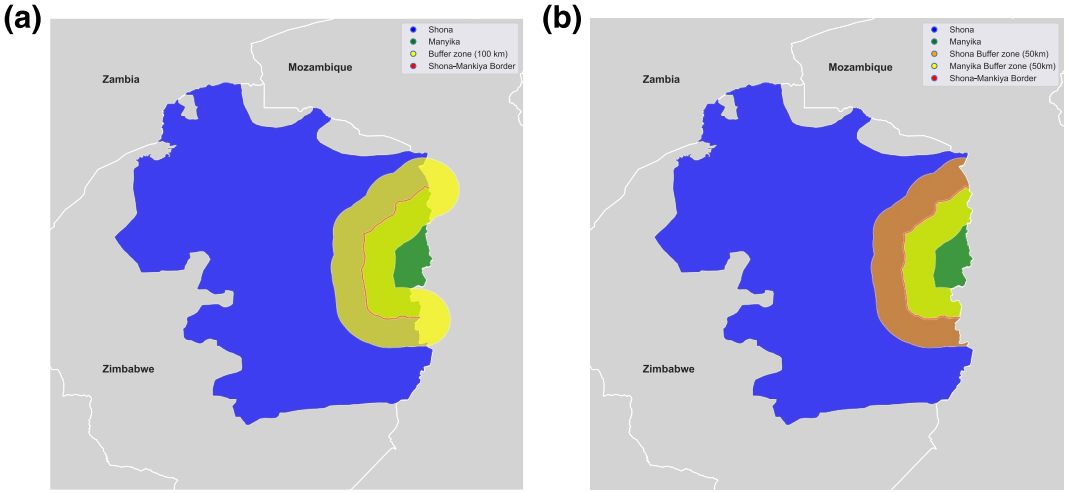

FIGURE 1 Buffer zones. (a) Original buffer zones (Replication, Ethnologue v16) (b) alternative buffer zones (Ethnologue v16). The figure shows two versions of buffer zones around the Shona-Manyika border in Zimbabwe. Panel (a) depicts the original buffer zones used in the replication, constructed using a 50 km radius around the border based on Ethnologue v16 data. Panel (b) presents alternative buffer zones, splitting the 100 km area into two distinct 50 km buffers for each group to capture potential differences in local conditions across ethnic groups. The Shona (blue) and Manyika (green) territories are highlighted, with each buffer zone shown in varying shades to distinguish the proximity to the ethnic boundary and the side of each border in which the buffer is located.

analyzes differences in a fundamental cultural trait—language. In particular, he explores whether linguistic distances - based on lexicographic distances (Wichmann et al., 2016) between spatially adjacent languages still spoken in the world in 2009 (Lewis, 2009)- were affected by potential gains from trade *between* ethnolinguistic groups. He proxies these potential gains using the standard deviation of agricultural productivity using the Caloric Suitability Index (CSI) in a 50 km buffer around the borders where their homelands meet (see Figure 1a).[1,2] Taking advantage of the spatial and temporal variation in agricultural productivity, he identifies the effect of the standard deviation of agricultural productivity on linguistic distances. Specifically, following Galor and Özak (2015, 2016), he exploits the quasi-random changes in productivity generated in the course of the Columbian Exchange (Crosby, 2003) as a natural experiment to identify their effect on linguistic distances, finding that larger standard deviations decreased linguistic distances between neighboring ethnolinguistic groups. While this main result is well identified and very robust, one major drawback of his analysis is his measure of potential gains from trade, which (as we show below) confounds *inter-ethnic* and *intra-ethnic* or more generalized trade.

To understand his hypothesis and the empirical issues better, consider two neighboring ethnolinguistic groups (like the Shona and Manyika depicted in Figure 1). Dickens (2022) argues through the lens of a Ricardian trade model, that if agricultural productivity differs *between* these two groups, then there will exist potential gains from *inter-ethnic* trade due to specialization. The larger these differences, the larger those potential gains, and, thus, more trade between these groups. So, the presence and frequency of inter-ethnic trade should have increased inter-ethnic contact and linguistic exchange, which, in turn, should have decreased linguistic differences between these two groups. In his empirical test, Dickens (2022) uses the standard deviation of agricultural productivity in the buffer around the border where these ethnic groups meet (yellow area in Figure 1a) as his proxy for potential for inter-ethnic trade. But, clearly, the potential for *inter-ethnic* trade should depend on the difference in productivity *between* ethnic groups around the border and not on the standard deviation of productivity in the buffer as a whole, that is, it should depend on the differences in agricultural productivity between the orange and yellow areas in Figure 1b.

In the present paper, we investigate whether his empirical results are reproducible and replicable, and further test his hypothesis using this alternative approach to measuring the potential gains from *inter-ethnic* trade. Our focus in this replication is less on the statistical robustness of the main results (which tends to be the focus of many replication exercises) or the reproducibility using the data and code produced by the author,[3] and more on the reproducibility and replicability of the empirical findings from original sources (which given the geographical nature of the analysis tends to be a more difficult task), and the empirical implementation of his theory. Specifically, (1) we reconstructed his main independent variables from original sources (Galor & Özak, 2015, 2016; Lewis, 2009), (2) we reconstructed his main independent variables using an updated version of the ethnolinguistic map he used to create the buffers in his analysis,

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TABLE 1 Reproduction of Table 1 in Dickens (2022) standardized betas.

	Dependent variable: Lexicostatistical linguistic distance ∈ (0, 1)					
	(1)	(2)	(3)	(4)	(5)	(6)
Δ in std CSI in buffer (post-1500)	−0.114	−0.113	−0.094	−0.117	−0.111	−0.078
	(0.034)	(0.038)	(0.038)	(0.038)	(0.037)	(0.041)
	[0.001]	[0.003]	[0.013]	[0.002]	[0.003]	[0.056]
std CSI in buffer (pre-1500)	−0.094	−0.093	−0.071	−0.066	−0.062	−0.046
	(0.033)	(0.034)	(0.040)	(0.033)	(0.039)	(0.042)
	[0.005]	[0.005]	[0.071]	[0.042]	[0.116]	[0.281]
Δ in average CSI in buffer (post-1500)		−0.003	0.002	−0.018	−0.004	0.027
		(0.036)	(0.035)	(0.036)	(0.035)	(0.037)
		[0.932]	[0.958]	[0.616]	[0.918]	[0.452]
Average CSI in buffer (pre-1500)		−0.005	−0.002	0.005	0.017	0.037
		(0.024)	(0.024)	(0.025)	(0.025)	(0.025)
		[0.847]	[0.927]	[0.846]	[0.481]	[0.134]
Geography controls	No	No	Yes	No	Yes	Yes
Spatial controls	No	No	No	Yes	Yes	Yes
Language family FEs	Yes	Yes	Yes	Yes	Yes	Yes
Country FEs	No	No	No	No	No	Yes
Adjusted R^2	0.248	0.248	0.259	0.264	0.278	0.369
Observations	8402	8402	8402	8402	8402	7291

Note: Unit of observation: border buffer zone (100 km). This table establishes the negative and statistically significant effect of variation in land productivity on a language pair's lexicostatistical linguistic distance. Geography controls include mean elevation, ruggedness, mean temperature and its standard deviation, mean precipitation and its standard deviation, and the prevalence of malaria. Spatial controls include logged distance to the nearest coast, country border, lake, major river and minor river, logged distance between group centroids, the absolute difference in latitude and longitude, logged land area, and logged population. SEs are double clustered at the level of each language group and are reported in parentheses.

(3) using the same original and updated sources, we constructed alternative (and perhaps more appropriate) measures of potential gains from inter-ethnic trade. In particular, instead of using the standard variation of agricultural productivity around the border, we proxy the potential for *inter-ethnic* trade by exploiting differences in agricultural productivity *across the border* (Figure 1b), which (as shown below) should better capture the gains from inter-ethnic trade. We further explore the role of inter-ethnic trade by analyzing the effect of cross-border differences in agricultural productivity in all crops underlying the CSI measures.

Using these various new measures, we successfully replicated his main result (column 5 of Table 1 in his paper), which we use for our analysis here. We provide a more complete replication in the repository with the new data and code. Our reconstructed independent variables, based on the same and updated underlying raw data, are very highly correlated with the ones provided in the replication package and generate almost identical results. Thus, we provide a successful direct replication of his work. Moreover, we find that our improved measures of potential gains from inter-ethnic trade generate weaker results in terms of sign and size of the coefficient of interest, while having almost identical explanatory power. Since our new measures capture mainly inter-ethnic trade potential, while the original measures reflect generalized or intra-ethnic trade, our results suggest that it is trade in general which matters for cultural convergence.

Besides contributing to our understanding of the role of trade in cultural convergence, our paper also contributes to the literature on replication in economics and other social sciences. In particular, we provide an open-source computational environment that can be used to replicate our and the original results of Dickens (2022), allowing the replicators to be replicated. Additionally, our tools and methods can be used to replicate other research, especially studies that employ geographical data or generate maps. Moreover, our tools can be used for teaching and learning purposes, especially for students, policy-makers, and other individuals in developing countries, who may not have

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access to proprietary software (ArcGIS, Stata, etc.) or sophisticated computational resources. In particular, to help others replicate and learn from our replication analysis, we have also created scripts to recreate our computational environment using conda/mamba, Deepnote, Docker, and MyBinder. All our data and code are available in the OPENICPSR repository https://doi.org/10.3886/E209325V2 and in OSF https://osf.io/k3p7g/.

2 | REPRODUCIBILITY

Dickens (2022) estimates the effect of his measures of potential gains from trade (standard deviation of agricultural productivity in the buffer around the common ethnolinguistic border) using a cross-section of all spatially adjacent language pairs and ordinary least squares regressions (his most comprehensive specification includes a large set of geographical and spatial controls, language family and country fixed effects). He double-clustered standard errors at the level of each language group to account for dependence between observations. His main results are presented in his Table 1 (p. 967). Specifically, in Column (1) he "reports within-family estimates for pre-Columbian land productivity variation and the post-Columbian change in land productivity variation. Both estimates enter with the expected negative sign and are statistically significant at the 1% level. This says that ethnic groups separated across high-variation regions are more similar in language than groups separated across low-variation regions. In particular, a one standard deviation increase in pre-1500 land productivity variation decreases linguistic distance by 1.7 percentage points, while a standard deviation increase in productivity variation at the onset of the Columbian exchanges implies a 2.0 percentage points decrease in linguistic distance." (p. 967) Additionally, he shows that after accounting for all controls except country fixed effects (column 5) "[t]he coefficient on pre-1500 land productivity variation retains the expected sign, but loses significance at standard levels. Whereas the coefficient of interest—the effect of land productivity variation resulting from the Columbian Exchange in the post-1500 period—retains statistical significance with the expected negative sign, and is statistically equivalent to the unconditional estimate in column (1)." (p. 968).

We downloaded the replication package provided by the author to the journal.[4] The code and data worked without any issues. We were able to reproduce the results of the paper using Stata-MP version 17. In order to improve reproducibility and provide educational tools based on these analyses for individuals who may not have access to this paid software, especially individuals in developing countries, we replicated these results using free tools. Specifically, we used the open-source programming language Python and various statistical and graphical packages based on it that are freely available.[5,6]

Our Table 1 reproduces the results of Table 1 in the original paper and shows the OLS estimates using standardized beta coefficients to simplify interpretability of the results (Table A1 in the Appendix reproduces the original Table 1 without this transformation) of the following equation:

$$\begin{aligned} LD_k = \beta_0 + \beta_{std}^{1500} CSIstd_k + \beta_{std}^{change} \Delta CSIstd_k \\ + \beta_{mean}^{1500} CSI_k + \beta_{mean}^{change} \Delta CSI_k + X_k' \Phi + \lambda_{l_i(k)} + \theta_{l_j(k)} + \delta_{c(k)} + \varepsilon_k, \end{aligned} \tag{1}$$

where LD_k is the linguistic distance in border k, $CSIstd_k$ is the standard deviation of CSI (pre-1500) and $\Delta CSIstd_k$ is its change (Δ) (post-1500) in buffer k, and CSI_k and ΔCSI_k are the average pre-1500 and its change post-1500 in CSI in buffer k. The results in column (5) of Table 1 suggest that after accounting for geographical and spatial controls, and language family fixed effects, a one standard deviation increase in the change in land productivity variation (post-1500) decreases linguistic distances by 0.11 standard deviations (p-value =0.003) and a one standard deviation increase in land productivity variation (pre-1500) decreases linguistic distances by 0.06 standard deviations (p-value =0.116). To simplify the presentation, we will focus on this same specification in the rest of the replication analysis.

3 | REPLICATION

3.1 | Direct replication using original and updated data

We also replicated the construction of the main independent variables used in his analysis. We assumed the original study employed the *langa.shp* file from the World Language Mapping System, which provides polygons for the

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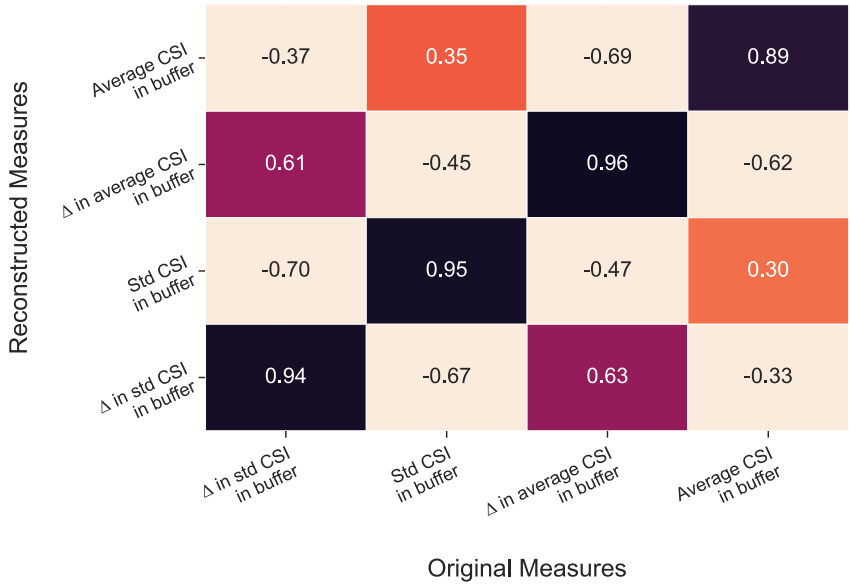

FIGURE 2 Correlation between original and reconstructed independent variables. The figure presents the correlation matrix between original and reconstructed independent variables for different measures of agricultural productivity based on CSI data within buffer zones. Each cell reports the correlation coefficient between the corresponding reconstructed and original variable based on Ethnologue (v16). Variables include the average CSI, change in average CSI, standard deviation of CSI, and change in standard deviation of CSI within the buffer. High correlations (close to 1 or −1) indicate a strong similarity between the original and reconstructed measures, while low correlations suggest distinct patterns in their behavior.

homelands of languages spoken across the world in 2009 (version 16) and 2014 (version 17) based on the Ethnologue (Lewis, 2009; Lewis et al., 2014). While the paper explains clearly most steps involved in the construction of the data, it did not clarify the geographical projection used in the analysis, which provided us with one degree of freedom. We followed the most straightforward (though perhaps not the most correct) strategy and used the original projections of the raw data in the construction.[7] Additionally, Galor and Özak (2015, 2016) provide two distinct Caloric Suitability Indices for the pre-1500 period, but Dickens (2022) does not clarify which one he used. Thus, we have a second degree of freedom, and we use their pre-1500 measure that includes Asian crops in Africa for our main replication analyses.[8]

Once we had these data sources and had made choices regarding the projections to be used, we followed a similar strategy to him to construct buffers around the borders. In the construction of borders and buffers, (i) following Dickens (2022), we excluded pairs in which the border reflected the same language in two countries (e.g., the language Abron is spoken both in Ghana and Côte d'Ivoire, Spanish in many neighboring Latin American countries, and similar for Arabic in the Middle East and North Africa). Clearly, including these would have been problematic and bias the analysis.[9] We also identified one language that had the same name (NAME2 in the original data), was spoken in two countries, but had different ISO codes. We decided to treat it as a unique language and exclude it from the data construction.[10] None of these language pairs were included in the (original) empirical analyses since they were excluded due to missing linguistic distance data. (ii) We excluded pairs in which the homeland of one language was (fully) contained inside the other, since this may reflect recent changes in the location of speakers and may bias the results.[11]

We reconstructed the data using both versions 16 and 17 of the Ethnologue. We found that between version 16 and 17, Ethnologue dropped many languages, which may have become extinct, or by mistake (e.g., version 17 does not have a polygon for Spanish in Paraguay). Additionally, some new languages appeared in the data set (not clear whether these are languages that did not have a polygon in version 16 or which became more vibrant). Specifically, version 17 has 414 languages not contained in version 16, while version 16 has 537 languages not contained in version 17. These differences explain the main differences in the samples for replication. In the main text, we focus on version 16, but all the results for version 17 are available in the appendix and replication code.

Our reconstructed independent variables, based on the same underlying raw data, are very highly correlated with the ones provided in the replication package, as shown in Figure 2. Similar high correlations exist between the original data and our reconstructed measures based on version 17 of the Ethnologue, as shown in Figure B1 in the Appendix. In

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TABLE 2 Replication and extension of main result.

	Dependent variable: Lexicostatistical linguistic distance ∈ (0, 1)						
	Replication					Robustness	
	Original data			Reconstructed data		Alternative measure	
	Original sample (1)	v16 (2)	v17 (3)	v16 (4)	v17 (5)	v16 (6)	v17 (7)
Δ in std CSI in buffer (post-1500)	−0.111	−0.111	−0.136	−0.140	−0.190		
	(0.037)	(0.037)	(0.040)	(0.042)	(0.042)		
	[0.003]	[0.003]	[0.001]	[0.001]	[0.000]		
Std CSI in buffer (pre-1500)	−0.062	−0.062	−0.076	−0.064	−0.105		
	(0.039)	(0.039)	(0.041)	(0.041)	(0.042)		
	[0.116]	[0.114]	[0.066]	[0.120]	[0.013]		
Δ in average CSI in buffer (post-1500)	−0.004	−0.003	0.017	0.007	0.039	−0.034	−0.022
	(0.035)	(0.035)	(0.035)	(0.043)	(0.043)	(0.031)	(0.031)
	[0.918]	[0.922]	[0.622]	[0.872]	[0.354]	[0.275]	[0.482]
Average CSI in buffer (pre-1500)	0.017	0.018	0.020	−0.003	0.012	0.008	0.006
	(0.025)	(0.025)	(0.026)	(0.032)	(0.033)	(0.024)	(0.025)
	[0.481]	[0.476]	[0.436]	[0.926]	[0.706]	[0.747]	[0.807]
Δ in cross border CSI difference (post-1500)						−0.049	−0.059
						(0.025)	(0.025)
						[0.049]	[0.016]
Cross border CSI difference (pre-1500)						0.001	−0.002
						(0.022)	(0.021)
						[0.965]	[0.935]
Geography controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Spatial controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Language family FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.278	0.278	0.280	0.279	0.282	0.277	0.279
Observations	8402	8400	7560	8400	7560	8400	7560

Note: Unit of observation: border buffer zone (100 km). This table establishes the negative and statistically significant effect of variation in land productivity on a language pair's lexicostatistical linguistic distance. Geography controls include mean elevation, ruggedness, mean temperature and its standard deviation, mean precipitation and its standard deviation, and the prevalence of malaria. Spatial controls include logged distance to the nearest coast, country border, lake, major river and minor river, logged distance between group centroids, the absolute difference in latitude and longitude, logged land area and logged population. SEs are double clustered at the level of each language group and are reported in parentheses.

Table 2, we explore how the reconstructed data and samples affect his main result from Table 1 column 5 in the original paper. Specifically, column (1) replicates the original result, while columns (2) and (3) show the result of constraining the original data to the samples of language pairs we have in the reconstructed data with qualitatively similar results, albeit a bit stronger effects in the reduced sample of version 17. Columns (4) and (5) replicate the analysis using our reconstructed measures and find almost identical results, although the estimated effects are always larger. Specifically, using the original data and sample, a one standard deviation increase in the change in land productivity variation (post-1500) decreased linguistic distances by 0.111 standard deviations (p-value =0.003), while it decreases it by 0.14 standard deviations (p-value =0.001) in the reconstructed Ethnologue version 16 data, and by 0.190 standard deviations (p-value <0.001) in the reconstructed Ethnologue version 17 data. Thus, there does not seem to exist significant differences

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between the replication and original analyses. This result is further confirmed in Figure B2a, which explores the effects of changing buffer sizes or accounting for country-fixed effects.

4 | ROBUSTNESS—INTER-ETHNIC VS GENERALIZED TRADE POTENTIAL

Dickens's (2022) "aim [...] is to shed light on an unexplored channel of cultural change: inter-ethnic trade" (p. 953). His main hypothesis, general theory, and model are based on the idea that "the gains from trade are a function of land productivity variation *between* groups" (p. 959, emphasis added). Thus, the idea is that linguistic distances should decrease due to *different* groups coming together to exchange *different* types of goods they specialize in.[12] Yet, neither his measure of inter-ethnic trade potential, the standard deviation of agricultural productivity, nor the evidence he presents based on it, *directly* provide support for this *interethnic* aspect of trade, since they fail to capture cross-ethnic economic specialization. In particular, the ethno-linguistic group level analyses, which show that a larger standard deviation of agricultural productivity is associated with more trade and less conflict (Section 4 of the paper, especially Table 4 and Figure 9), is similar and in line with previous research that has established the positive effect of diversity (ecological, geographical, and population) within an ethno-linguistic group (or its homeland) on the emergence of economic specialization, trade, and states (Depetris-Chauvin & Özak, 2020b; Fenske, 2014), which can directly reduce linguistic distances across languages in a region. Thus, his border level analyses may reflect the negative effect of *generalized* trade on linguistic distances even in cases where there is no cross-ethnic specialization. Specifically, it may very well be the case that linguistic distances decrease because people from *different* ethnic groups interact when trading *similar* goods, that is, these interactions are not driven by ethnic level specialization. While this would have no impact on the empirical analyses themselves, as agricultural variation would decrease linguistic distances, it would affect their interpretation.

To understand this concern better, Figure 3 shows that the potential for *inter-ethnic* trade should depend (as in his model and theory) on the difference in productivity *between* ethnic groups around the border and not on the variation of productivity in the buffer as a whole, which should reflect incentives for generalized trade.[13] In particular, these figures depict two artificial buffers around an ethnic border, which is assumed to have the same mean and standard deviation in productivity, that is, they would have the same trade potential according to the original measure. But, clearly, inter-ethnic trade potential should be higher in Figure 3b than in Figure 3a since it provides stronger incentives for cross-ethnic economic specialization.

To explore this issue further, as a final step, we construct measures of the difference in productivity pre-1500 and its change post-1500 between the two sides of each buffer (as depicted in Figure 1b) and replicate the main specification using these. Specifically, for each buffer, we compute the absolute difference in productivity pre-1500 between both sides of the border, that is, for each part of the buffer that belongs to a specific homeland, and the change in this

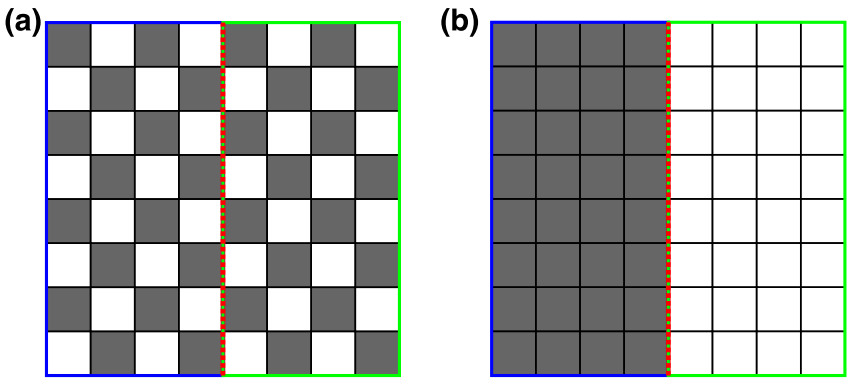

FIGURE 3 Productivity and inter-ethnic trade potential in buffer around border. (a) Similar productivity and low potential (b) different productivity and high potential. The figure illustrates how inter-ethnic trade potential is influenced by productivity differences *between* ethnic groups at the border, rather than by overall productivity variation within the buffer region. Both panels have the same mean and standard deviation of productivity, yet trade potential differs based on inter-ethnic productivity disparity. Panel (a) shows similar productivity levels on both sides of the border, implying low trade potential. In contrast, Panel (b) depicts distinct productivity levels across the border, suggesting higher trade potential.

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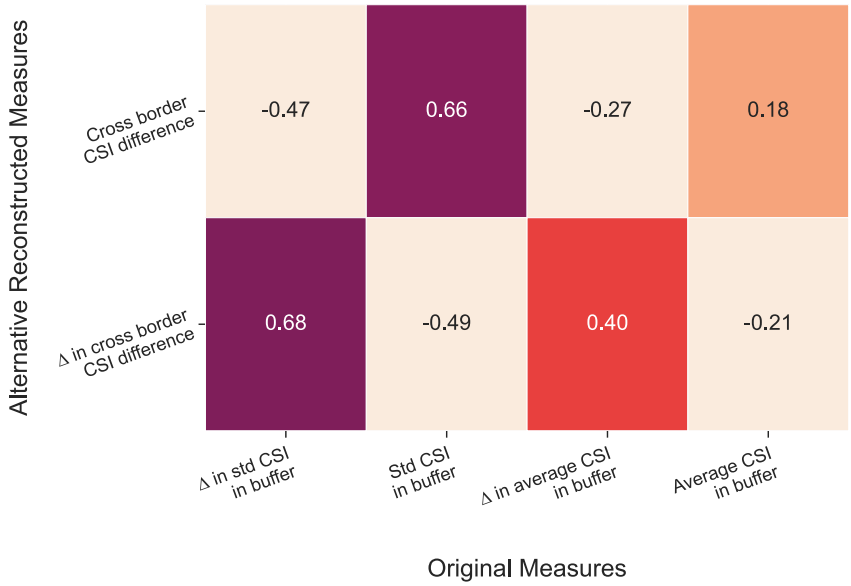

FIGURE 4 Correlation between original and alternative measures of potential gains from inter-ethnic trade based on Ethnologue v16. The figure presents the correlation matrix between original and alternative measures of potential gains from inter-ethnic trade based on Ethnologue (v16) data. Each cell shows the correlation coefficient between the corresponding original and alternative measure. Variables include cross-border CSI differences, changes in cross-border CSI differences, as well as within-buffer average CSI, changes in average CSI, standard deviation of CSI, and changes in standard deviation of CSI. High correlations indicate similarity between the original and alternative measures, while low correlations reflect significant differences.

difference post-1500. For example, given the border between ethnicities A and B, our new measure is $\left|CSI_k^A - CSI_k^B\right|$ where CSI_k^i is the average CSI on side $i = A, B$ of the buffer. We similarly compute its change due to the Columbian Exchange. Using these measures, we estimate the following regression

$$\begin{aligned} LD_k = \beta_0 + \beta_{alt}^{1500} CrossBorder_k + \beta_{alt}^{change} \Delta CrossBorder_k \\ + \beta_{or}^{1500} CSI_k + \beta_{or}^{change} \Delta CSI_k + X_k' \Phi + \lambda_{l_i(k)} + \theta_{l_j(k)} + \varepsilon_k, n \end{aligned} \tag{2}$$

where $CrossBorder_k$ and $\Delta CrossBorder_k$ are the new measures for border k. Clearly, a significant effect of these differences, that is, $\beta_{alt}^{1500} < 0$ and $\beta_{alt}^{change} < 0$, would be prima facie evidence for the role of inter-ethnic trade. Columns (6) and (7) in Table 2 show the results of this analysis, in which we replace the original proxies of inter-ethnic trade (standard deviation of agricultural productivity and its change) with our new measures. Specifically, the results show the effect of differences in productivity across the border pre-1500 and its change post-1500 on linguistic distances after accounting for all the same controls as in the main specification we have been focusing on in this replication (i.e., Column (5) in Table 1 of the original study). The estimates suggest that a one standard deviation increase in the changes post-1500 in the differences in agricultural productivity across the two sides of the borders decreases linguistic distances by 0.049 (p-value $= 0.049$) and by 0.059 (p-value $= 0.016$) using data from Ethnologue version 16 and 17 respectively. This is strong direct evidence that inter-ethnic trade does affect linguistic distances.

Still, the results may reflect the forces of generalized trade. In particular, as Figures 4 and B3 show, the correlation between our alternative measures of inter-ethnic trade potential based on cross-border differences and the original measures of agricultural productivity variation both in the pre- and post-1500 periods is quite high with correlations above 0.65.[14] Clearly, given these high correlations, disentangling the two forces is difficult.[15] To further explore this issue, we construct crop-specific measures of inter-ethnic trade potential. Specifically, for all crops in Galor and Özak (2016, 2015), we constructed measures of the cross-border difference in crop productivity (similar to our alternative proxy based on CSI). Unlike the original measures, we do not have changes, since the changes were based on the introduction of new crops. Using these measures, we reestimated Equation (2) using crop-specific instead of CSI cross-border differences. The results are shown in Figure 5. In particular, in Figure 5a, we plot the coefficient and standard error when we include each crop individually in Equation (2), that is, it plots the coefficients of 40 individual

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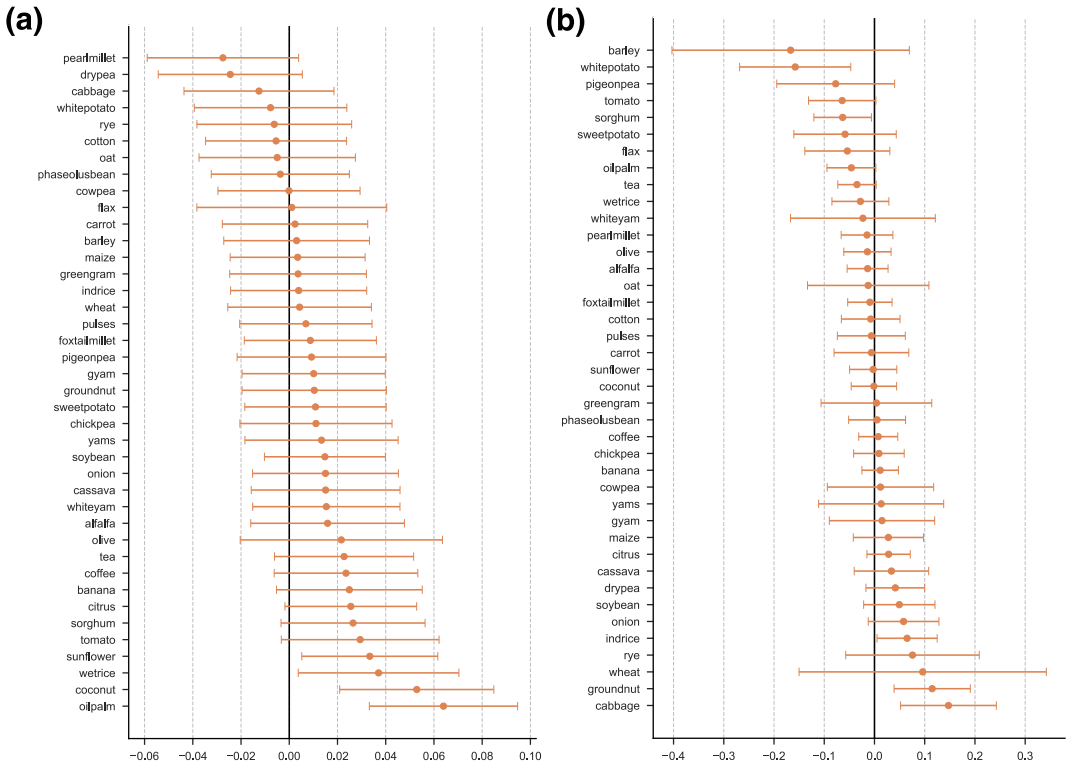

FIGURE 5 Coefficient plots—crop specific estimations. (a) Cross border difference (individual regressions) (b) cross border difference (joint regression). The figure displays coefficient plots for crop-specific estimations using cross-border differences. Panel (a) shows the results from individual regressions for each crop, while Panel (b) presents the results from a joint regression including all crops. Each coefficient estimate is represented by a dot, and the horizontal lines indicate the 95% confidence intervals. Positive coefficients suggest that higher productivity differences are associated with greater linguistic distances, while negative coefficients imply that higher productivity differences are linked to smaller linguistic distances. The variation in coefficient estimates across crops highlights the heterogeneous effects of cross-border differences on productivity.

regressions. It shows that most coefficients are not statistically different from zero (except sunflower, wet rice, coconut, and oil palm, which are significant, but have the wrong sign). In Figure 5b, we plot the coefficients when we include *all* crops jointly with similar results. In this case, white potato and sorghum have negative and statistically significant coefficients, while groundnut and cabbage have positive and statistically significant coefficients.

Thus, the results fail to provide strong evidence in favor of the inter-ethnic aspect of trade and may reflect the role of generalized trade instead. Nonetheless, these results need to be taken with a grain of salt, since there are many issues that may still bias the analysis. In particular, the location of these contemporary linguistic borders may not reflect historical ethnic borders (Depetris-Chauvin & Özak, 2020a), so that our measures may reflect poorly the historical conditions that may have generated inter-ethnic specialization. Moreover, our measures may not fully capture the incentives to specialize, for example, across crops (see, e.g., Blouin & Dyer, 2021). Additionally, it is important to remember, that even if endowments are similar, inter-ethnic specialization and trade may still have evolved and driven linguistic convergence. Clearly, more research is needed.

5 | CONCLUSION

Dickens (2022) presents novel and important evidence to understand the cultural convergence between ethno-linguistic groups and the role trade and geography play in it. Following his methodology, we have successfully replicated his central result establishing the negative effect of the standard deviation of agricultural productivity on linguistic distances. We have shown the results are robust to changes in the construction method and the underlying linguistic maps finding that the empirical analyses remained qualitatively unchanged, although the estimated effects tended to be larger.

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As an extension of his work, we have delved deeper into the role of inter-ethnic as opposed to generalized trade in this result. Since both types of trade should increase contact between individuals, they should decrease linguistic distances due to increased contact among individuals. We argue that the original measures used in the study failed to properly distinguish between these two forces and construct novel measures of potential gains from inter-ethnic trade at ethno-linguistic borders. Using our novel measures, we obtained qualitatively similar although quantitatively smaller effects. Also, we find that these results are not robust to accounting for the original measures. This may reflect the fact that it is generalized trade and not inter-ethnic trade that drives linguistic convergence, or various shortcomings of the new measures. Fully understanding the role of inter-ethnic trade may require different strategies like building more sophisticated measures that account for potential mismeasurement issues (Depetris-Chauvin & Özak, 2020a) or proper crop specialization across ethnic groups (Blouin & Dyer, 2021).[16] Clearly, more research is needed to understand this important subject.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in OPENICPSR at https://doi.org/10.3886/E209325V2, reference number openicpsr-209325 (Gonzalez & Özak, 2024).

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ENDNOTES

[1] That is, buffers with a diameter of 100 km.

[2] The Caloric Suitability Indices (CSI) capture the variation in potential crop yield across the globe, as measured in calories per hectare per year. Moreover, in light of the expansion in the set of crops that are available for cultivation in the course of the Columbian Exchange, the CSI indices provide a distinct measure for caloric suitability for the pre-1500 and the post-1500 era. They are based on a set of 48 crops and data from the GAEZ project (see Galor & Özak, 2015, 2016).

[3] Both us and, independently, the Institute for Replication were successfully able to reproduce the original study using the replication package provided by the author on the journal's website. We did not find any differences between the published results and the ones we obtained using his data and code. The replication code and data is available from the website of the Economic Journal (https://doi.org/10.1093/ej/ueab065). The code reproduces all tables and most figures (except maps), although it does not save the output to a usable file.

[4] Downloaded from https://doi.org/10.1093/ej/ueab065 on October 4, 2022.

[5] Our reproduction uses Jupyter notebooks and performs the same analyses in Python and Stata. To execute cells in the notebooks that use Stata, this software needs to be installed on the computer.

[6] We also used Julia, another open-access programming language, to replicate Dicken's (2022) tables and figures.

[7] Both the linguistic maps for Ethnologue (Lewis, 2009; Lewis et al., 2014) and the Caloric Suitability Index (Galor & Özak, 2015, 2016) are provided in the WGS84, that is, EPSG:4326, geographic coordinate system. These are not the best projections to use in a global analysis given the underlying data and question. Using a projected coordinate system like a cylindrical equal area projection (ESRI:54034) would have been preferable. Yet, previous experience suggests that using either of these generates highly correlated measures so we do not consider this a major issue. Moreover, it provides us with an alternative robustness analysis, which clearly could be improved by selecting different projections in the construction.

[8] Our replication code and data reconstructs the data using all versions of the CSI data.

[9] In the original replication file, we found 111 pairs of languages that were the same language across two different countries (Stata command: tab identifier if langIso1 = = langIso2). Yet, these pairs are never used in the analyses in the original paper.

[10] The language is Marwari which is spoken in India and Pakistan (see notebook for references).

[11] Our sample using version 16 of the Ethnologue map differed from the original study by only 1 observation, which is the pair BZX-MLI-FFM-MLI (Bozo, Hainyaxo and Fulfulde, Maasina in Mali), where one is strictly contained in the other. The replication data contains a variable called overlap, which is used in Table A2 in the original paper to explore the robustness to excluding languages that overlap. We assumed these would include the ones we excluded due to being one contained in the other, but this variable identifies many more language pairs as overlapping. We find that 2848 of the language pairs with linguistic distance data that are identified as overlapping according to the original study are not identified as overlapping in our procedure. These include cases like KHK-MNG-CMN-CHN, AAC-PNG-DBY-PNG, and AAL-CMR-KOT-CMR, which the Ethnologue map does not show as overlapping. Without the code used to generate

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the original data we could not determine how these overlaps were identified and why our results differed in this aspect. We did not explore this issue further since it had no impact on the main regression we analyze here.

[12] One aspect that is not studied here nor in the original paper is how those differences come about. Specifically, whether the convergence is due to horizontal or vertical transmission of culture, and whether the type of trade can affect transmission via one channel or the other. We thank referee 2 for this insight.

[13] Equivalently, if one were to do the analysis using the complete homelands, it should be driven by the difference in productivity between the homelands and not by the variations across both homelands.

[14] This could be expected due to the way the data is constructed and the relation between the standard deviation across the whole buffer and difference in the mean across both sides of the border.

[15] For example, in a horse race, the effect of our measures of inter-ethnic trade potential fall between 77% and 80% and become statistically insignificant with p-values over 0.6. On the other hand, the original measures, which reflect generalized trade, remain basically unchanged and remain statistically significant at the 1% level.

[16] It should not be difficult to use the code we provide in our repository to extend these analyses further.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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APPENDIX A

TABLE A1 Reproduction of Table 1 in Dickens (2022).

	Dependent variable: Lexicostatistical linguistic distance ∈ (0, 1)					
	(1)	(2)	(3)	(4)	(5)	(6)
Δ in std CSI in buffer (post-1500)	−0.100	−0.100	−0.083	−0.103	−0.098	−0.065
	(0.030)	(0.034)	(0.034)	(0.034)	(0.033)	(0.034)
	[0.001]	[0.003]	[0.013]	[0.002]	[0.003]	[0.056]
std CSI in buffer (pre-1500)	−0.061	−0.061	−0.046	−0.043	−0.040	−0.029
	(0.022)	(0.022)	(0.026)	(0.021)	(0.026)	(0.027)
	[0.005]	[0.005]	[0.071]	[0.042]	[0.116]	[0.281]
Δ in average CSI in buffer (post-1500)		−0.001	0.001	−0.005	−0.001	0.008
		(0.010)	(0.010)	(0.010)	(0.010)	(0.010)
		[0.932]	[0.958]	[0.616]	[0.918]	[0.452]
Average CSI in buffer (pre-1500)		−0.001	−0.001	0.001	0.004	0.009
		(0.006)	(0.006)	(0.006)	(0.006)	(0.006)
		[0.847]	[0.927]	[0.846]	[0.481]	[0.134]
Geography controls	No	No	Yes	No	Yes	Yes
Spatial controls	No	No	No	Yes	Yes	Yes
Language family FEs	Yes	Yes	Yes	Yes	Yes	Yes
Country FEs	No	No	No	No	No	Yes
Adjusted R^2	0.248	0.248	0.259	0.264	0.278	0.369
Observations	8402	8402	8402	8402	8402	7291

Note: Unit of observation: border buffer zone (100 km). This table establishes the negative and statistically significant effect of variation in land productivity on a language pair's lexicostatistical linguistic distance. Geography controls include mean elevation, ruggedness, mean temperature and its standard deviation, mean precipitation and its standard deviation, and the prevalence of malaria. Spatial controls include logged distance to the nearest coast, country border, lake, major river and minor river, logged distance between group centroids, the absolute difference in latitude and longitude, logged land area and logged population. SEs are double clustered at the level of each language group and are reported in parentheses.

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TABLE A2 Replication and extension of main result non-standardized coefficients.

	Dependent variable: Lexicostatistical linguistic distance ∈ (0, 1)						
	Replication					Robustness	
	Original data			Reconstructed data		Alternative measure	
	Original sample (1)	v16 (2)	v17 (3)	v16 (4)	v17 (5)	v16 (6)	v17 (7)
Δ in std CSI in buffer (post-1500)	−0.111	−0.111	−0.136	−0.140	−0.190		
	(0.037)	(0.037)	(0.040)	(0.042)	(0.042)		
	[0.003]	[0.003]	[0.001]	[0.001]	[0.000]		
Std CSI in buffer (pre-1500)	−0.062	−0.062	−0.076	−0.064	−0.105		
	(0.039)	(0.039)	(0.041)	(0.041)	(0.042)		
	[0.116]	[0.114]	[0.066]	[0.120]	[0.013]		
Δ in average CSI in buffer (post-1500)	−0.004	−0.003	0.017	0.007	0.039	−0.034	−0.022
	(0.035)	(0.035)	(0.035)	(0.043)	(0.043)	(0.031)	(0.031)
	[0.918]	[0.922]	[0.622]	[0.872]	[0.354]	[0.275]	[0.482]
Average CSI in buffer (pre-1500)	0.017	0.018	0.020	−0.003	0.012	0.008	0.006
	(0.025)	(0.025)	(0.026)	(0.032)	(0.033)	(0.024)	(0.025)
	[0.481]	[0.476]	[0.436]	[0.926]	[0.706]	[0.747]	[0.807]
Δ in cross border CSI difference (post-1500)						−0.049	−0.059
						(0.025)	(0.025)
						[0.049]	[0.016]
Cross border CSI difference (pre-1500)						0.001	−0.002
						(0.022)	(0.021)
						[0.965]	[0.935]
Geography controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Spatial controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Language family FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.278	0.278	0.280	0.279	0.282	0.277	0.279
Observations	8402	8400	7560	8400	7560	8400	7560

Note: Unit of observation: border buffer zone (100 km). This table establishes the negative and statistically significant effect of variation in land productivity on a language pair's lexicostatistical linguistic distance. Geography controls include mean elevation, ruggedness, mean temperature and its standard deviation, mean precipitation and its standard deviation, and the prevalence of malaria. Spatial controls include logged distance to the nearest coast, country border, lake, major river and minor river, logged distance between group centroids, the absolute difference in latitude and longitude, logged land area, and logged population. SEs are double clustered at the level of each language group and are reported in parentheses.

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APPENDIX B

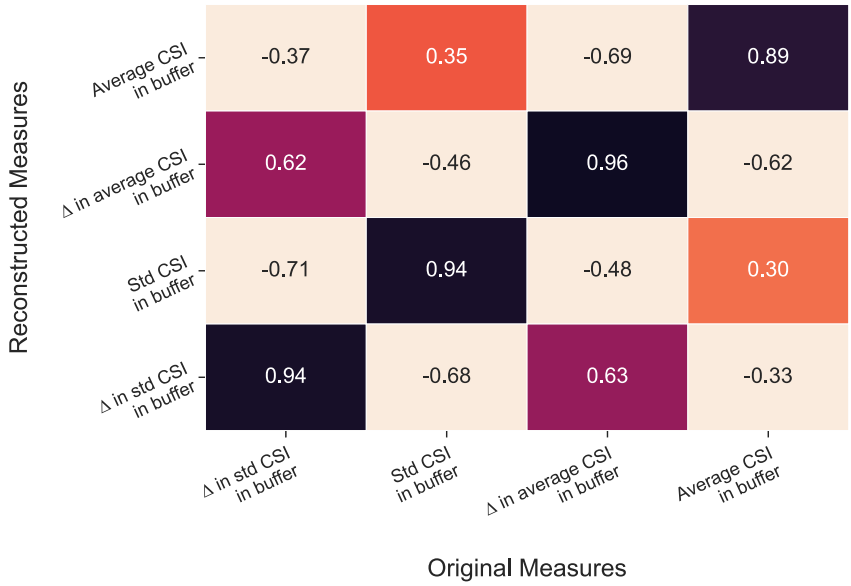

FIGURE B1 Correlation between original and reconstructed independent variables—Ethnologue v17. The figure presents the correlation matrix between original and reconstructed independent variables for different measures of agricultural productivity based on CSI data within buffer zones. Each cell reports the correlation coefficient between the corresponding reconstructed and original variable based on different versions of the Ethnologue (v16 and v17). Variables include the average CSI, change in average CSI, standard deviation of CSI, and change in standard deviation of CSI within the buffer. High correlations (close to 1 or −1) indicate a strong similarity between the original and reconstructed measures, while low correlations suggest distinct patterns in their behavior.

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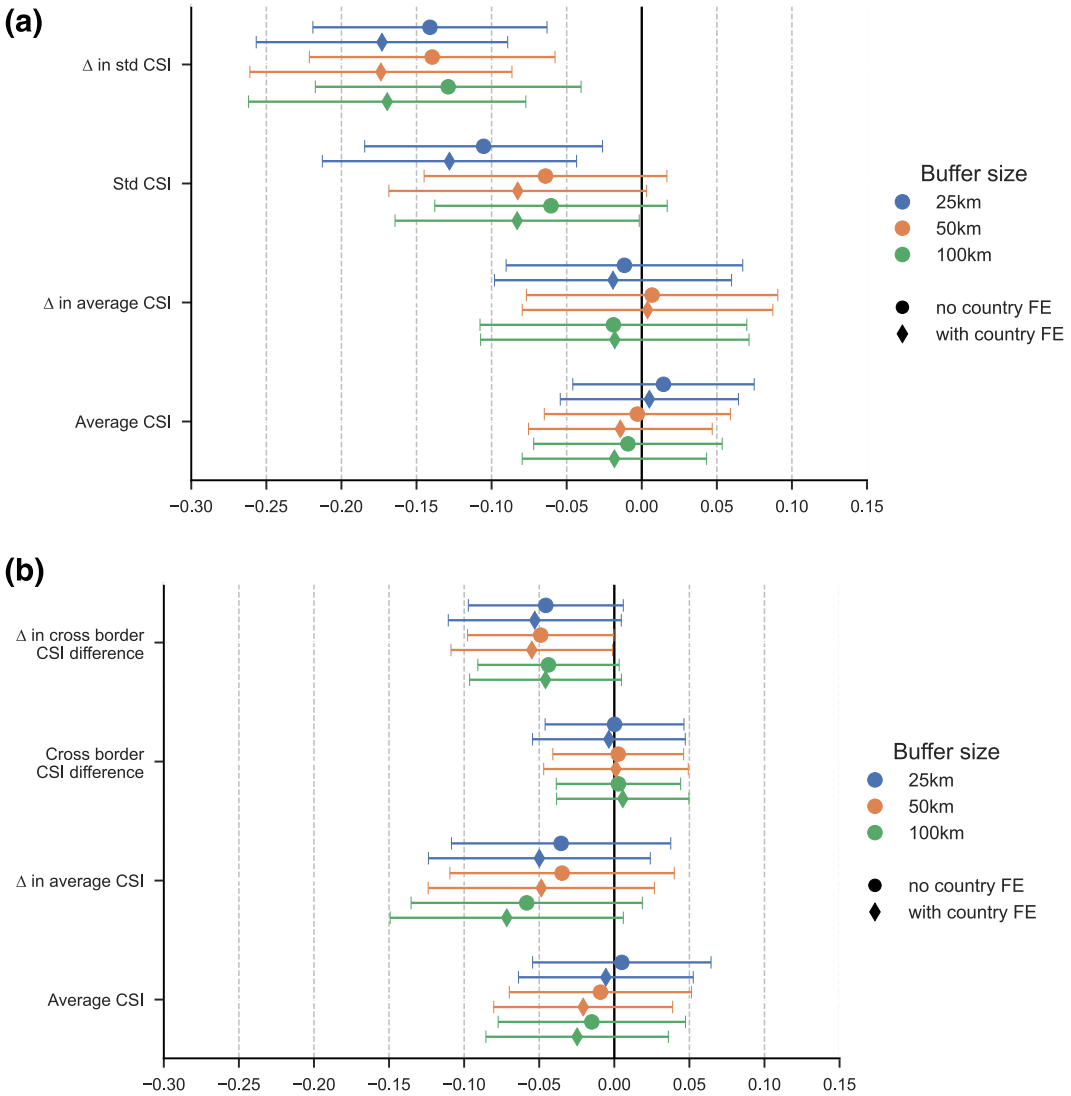

FIGURE B2 Replication and extension of main result—robustness to buffer sizes. (a) Original measure. (b) Alternative measure. The figure examines the robustness of the main results from Table 2 to variations in buffer sizes and the inclusion of country fixed effects. Panel (a) displays the OLS coefficients for the original measure, corresponding to the specification in column (4) of Table 2, while Panel (b) shows the coefficients for the alternative measure, replicating column (6) of Table 2. Different colors represent buffer sizes of 25 km, 50 km, and 100 km. Circles denote estimates without country fixed effects, and diamonds indicate estimates with country fixed effects. Horizontal bars show the 95% confidence intervals.

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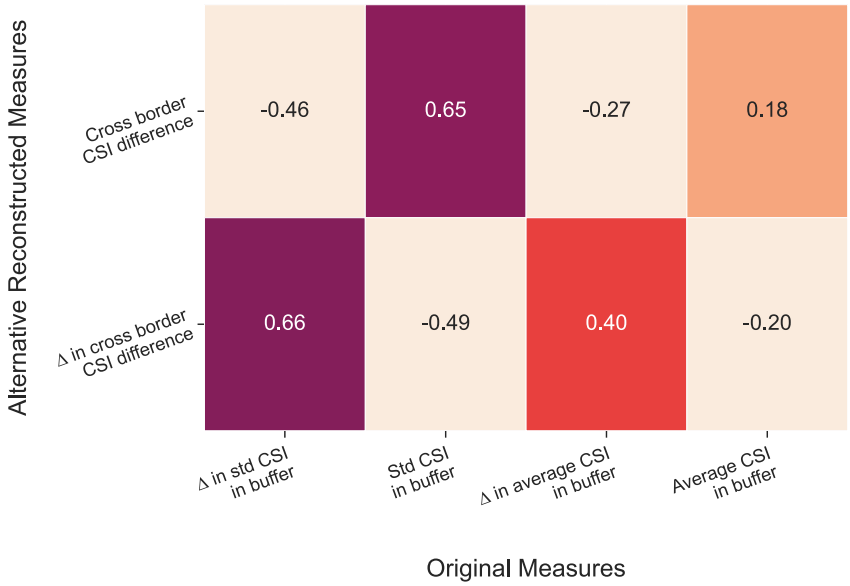

FIGURE B3 Correlation between original and alternative measures of potential gains from inter-ethnic trade—based on Ethnologue v17. The figure presents the correlation matrix between original and alternative measures of potential gains from inter-ethnic trade based on Ethnologue (v16 and v17) data. Each cell shows the correlation coefficient between the corresponding original and alternative measure. Variables include cross-border CSI differences, changes in cross-border CSI differences, as well as within-buffer average CSI, changes in average CSI, standard deviation of CSI, and changes in standard deviation of CSI. High correlations indicate similarity between the original and alternative measures, while low correlations reflect significant differences.

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